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## 1.Amazon EBS (Elastic Block Store)

**What is Amazon EBS?**
Amazon Elastic Block Store (EBS) is a **block-level storage service** provided by Amazon Web Services that is used with EC2 instances.
It provides **persistent storage**, which means data remains even after the EC2 instance is stopped or terminated.

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**Key Characteristics of EBS**

- Block-level storage
- High performance and low latency
- Persistent storage
- Attached to a single EC2 instance at a time

**Limitations of EBS**

- Works within same Availability Zone
- Cannot be accessed directly from internet

## 2.Amazon S3 (Simple Storage Service)

**What is Amazon S3?**
Amazon Simple Storage Service (S3) is an **object storage service** that stores data in the form of objects inside buckets.
It is designed to store and retrieve any amount of data from anywhere.

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**Key Characteristics of S3**

- Object-based storage
- Highly durable and scalable
- Accessible via internet
- Stores files as objects inside buckets

## 1. Cinder (Block Storage Service)

- **Cinder provides Block Storage**, similar to a virtual hard disk.
- **What it does**
  - Creates **volumes** (virtual disks)
  - Attaches volumes to virtual machines
  - Used for OS disks and application data
- **Characteristics**
  - Acts like a raw disk
  - Needs a filesystem (ext4, xfs, etc.)
  - Low latency
  - Persistent storage

- **Example Use Cases**
  - VM root disk
  - Database storage
  - Application storage
- **Simple Analogy**

➞ Like an **external hard drive** attached to your computer

## 2.Swift (Object Storage Service)

- **Swift provides Object Storage**, similar to Amazon S3.
- **What it does**
  - Stores data as **objects** (file + metadata)
  - Accessed using REST APIs / HTTP
  - Highly scalable
- **Characteristics**
  - No mounting like disk
  - No filesystem needed
  - Stores files, images, videos, backups
- **Example Use Cases**
  - Image storage
  - Video storage
  - Backup and archive
  - Static website files
- **Simple Analogy**

➞ Like **Google Drive / Dropbox**

**Cinder => Disk for VM**
**Swift=> Storage for files**

## To create an Amazon S3 bucket and deploy a static website using AWS S3 static website hosting.

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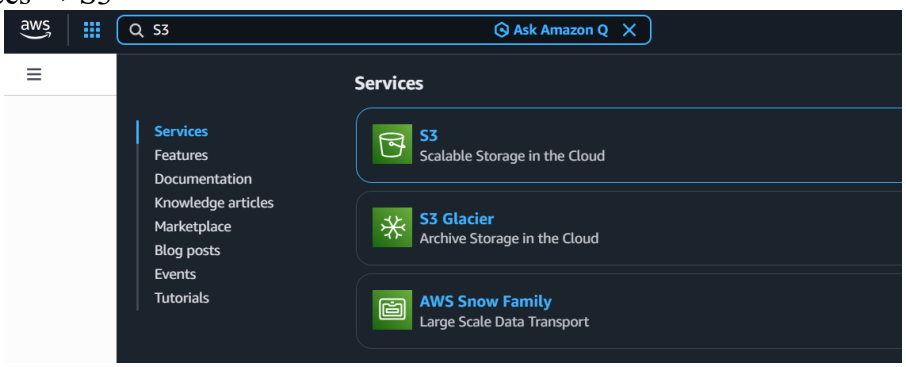
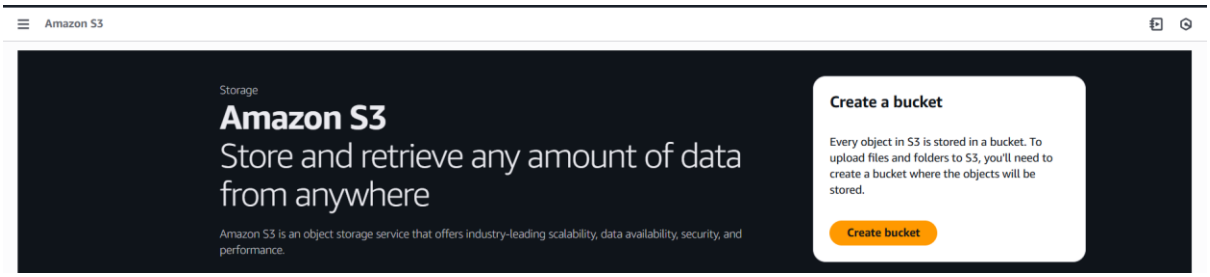
**STEP 1: Login to AWS Console**
**STEP 2: Open S3 Service**
Go to:

Services → S3



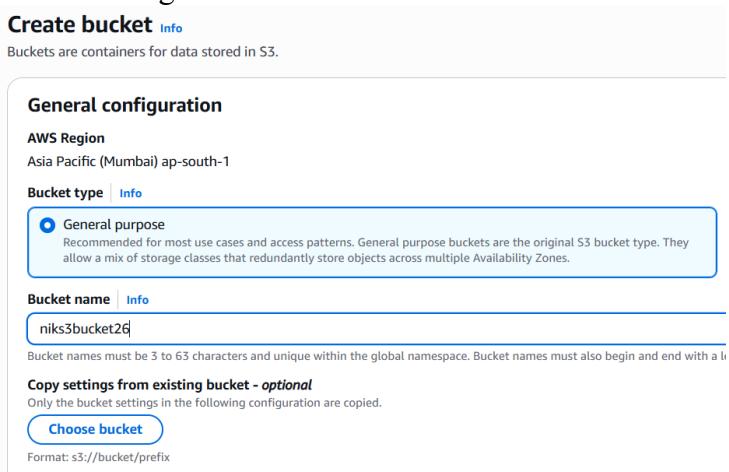
## STEP 3: Create S3 Bucket

Click: Create bucket



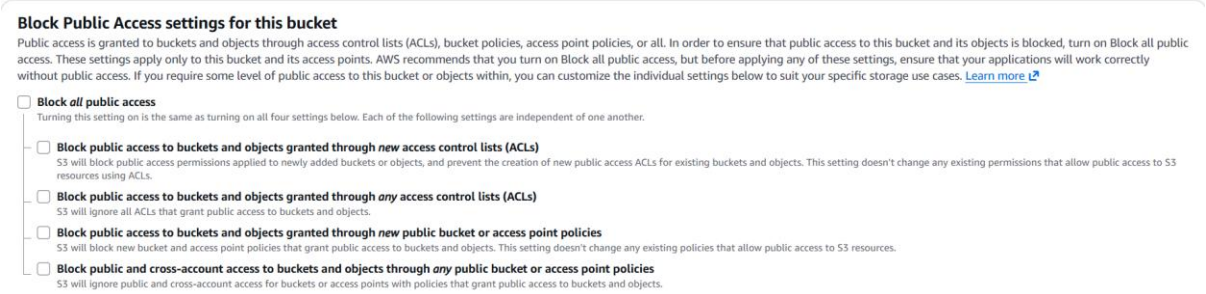

## STEP 4: Enter Bucket Name & Region

- Bucket name → must be unique
- Region → Select nearest region

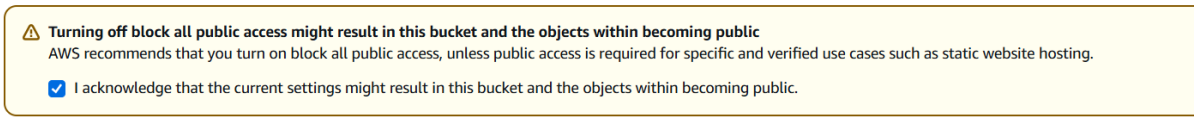



## STEP 5: Disable Block Public Access

Uncheck: Block all public access

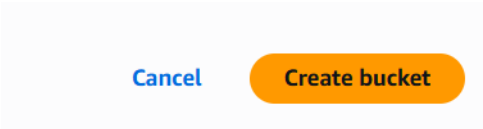



Tick:

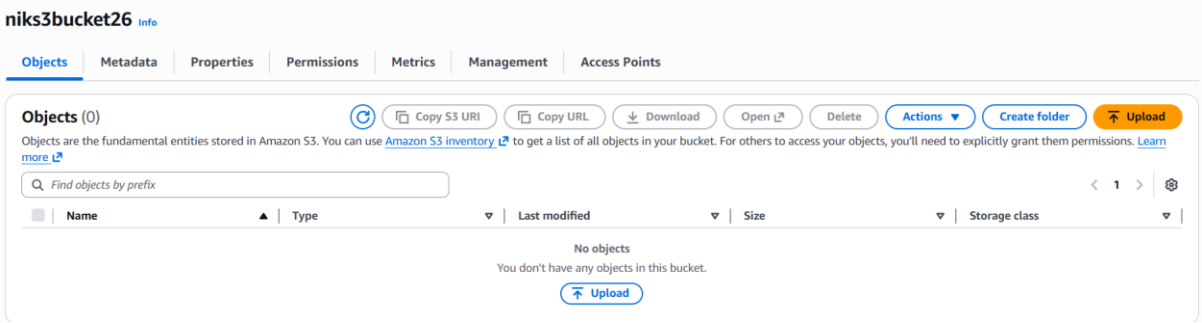



## STEP 6: Create Bucket

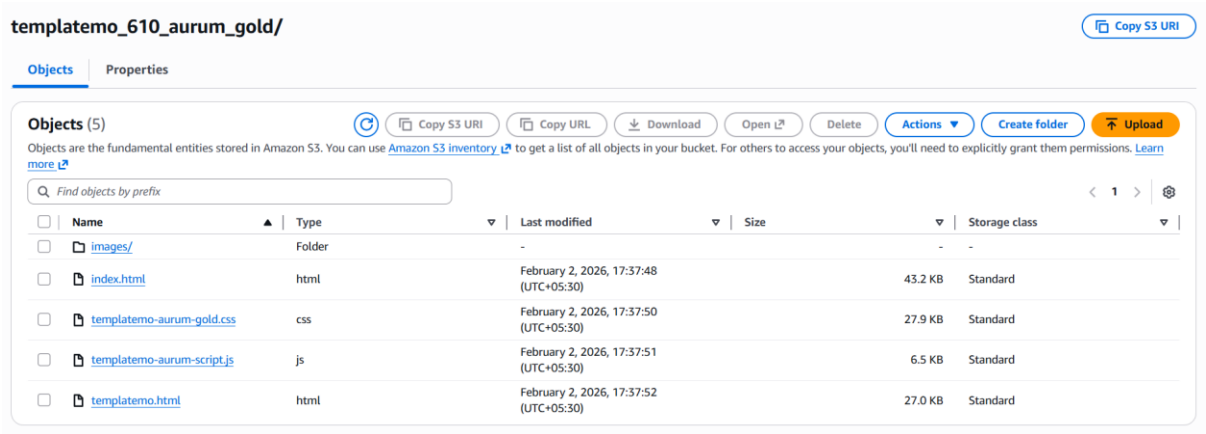
Click: Create bucket



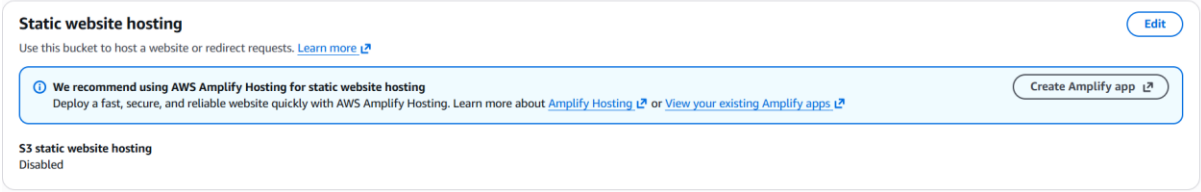
**STEP 7: Open Bucket**
Click on your bucket name.

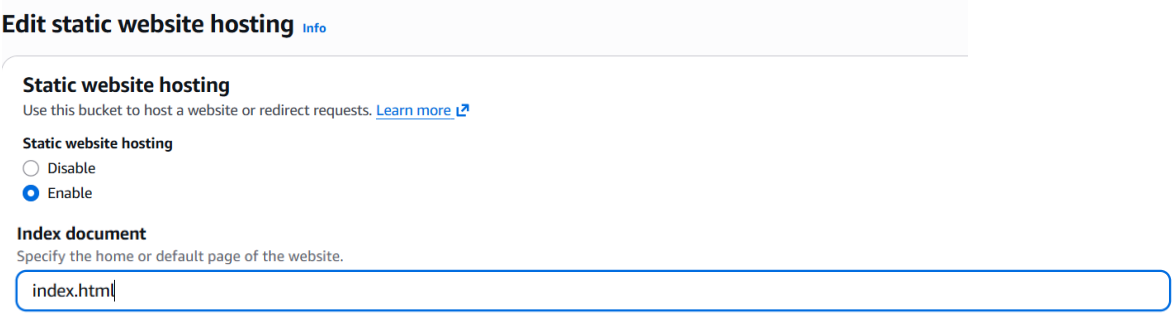



**STEP 8: Upload Website Files**
Click:
Upload → Add files → Add folder → Upload




**STEP 9: Enable Static Website Hosting**
Go to:
Properties → Static website hosting → Edit



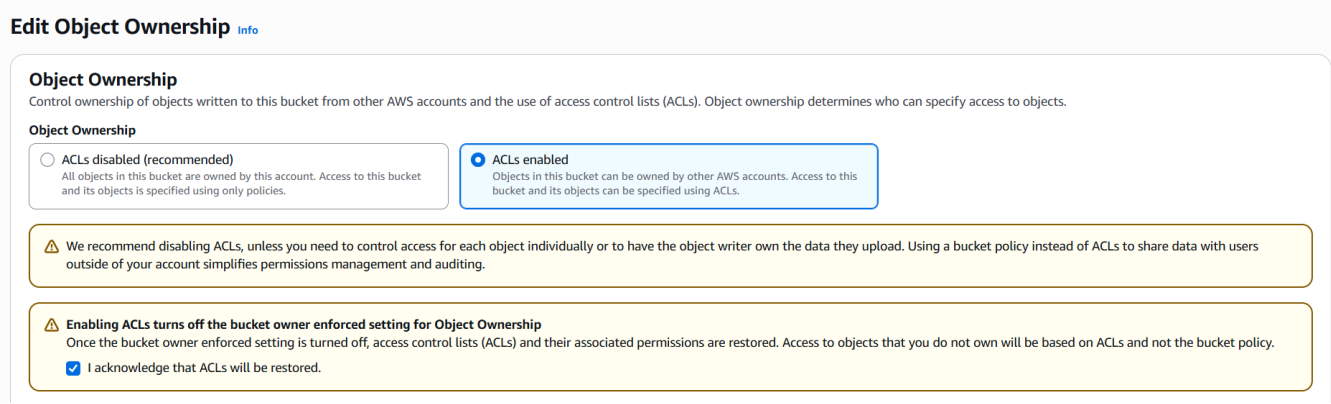




Select:

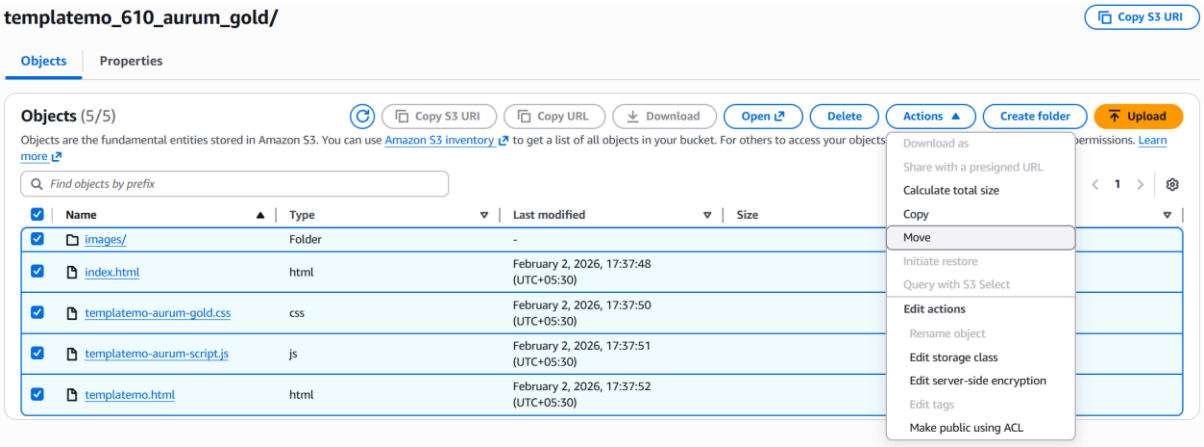
Enable
Hosting type → Static website hosting
Index document → index.html

**Save changes.**

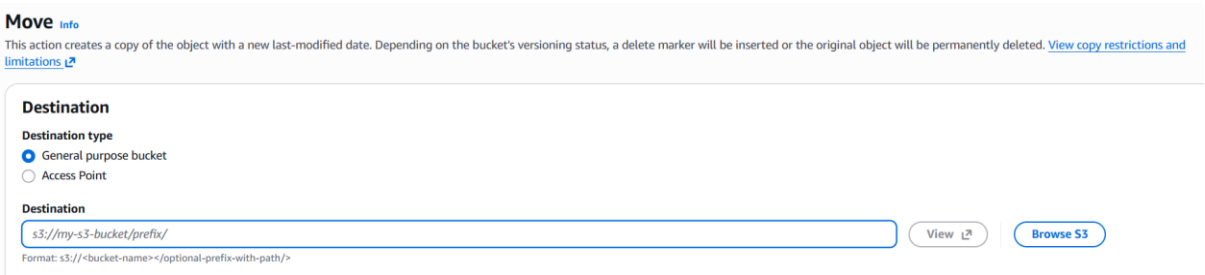
**STEP 10: Enable Access control lists**
Go to:
Permissions → Object Ownership→ Edit



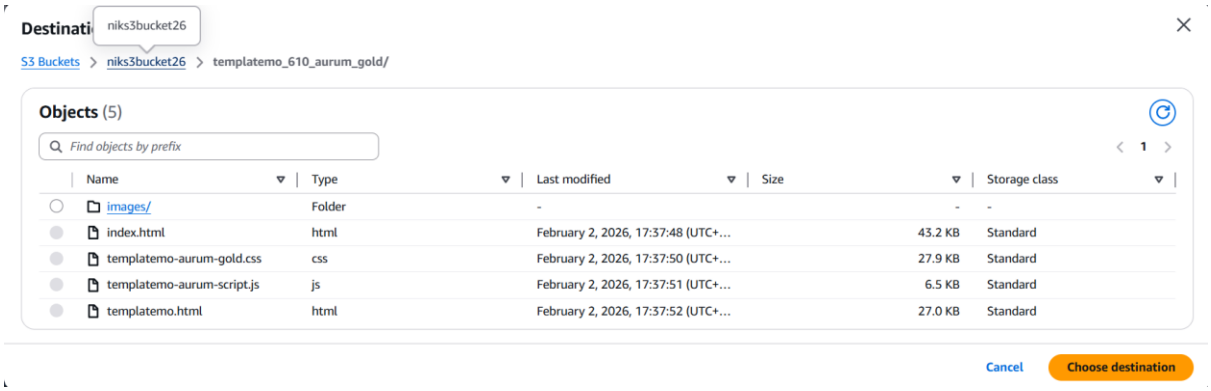

**STEP 11: Select Website Files to Move**
Inside the S3 bucket, select the website files and folders (such as **index.html**, CSS, JS, images folder) that need to be organized or moved to a specific directory.



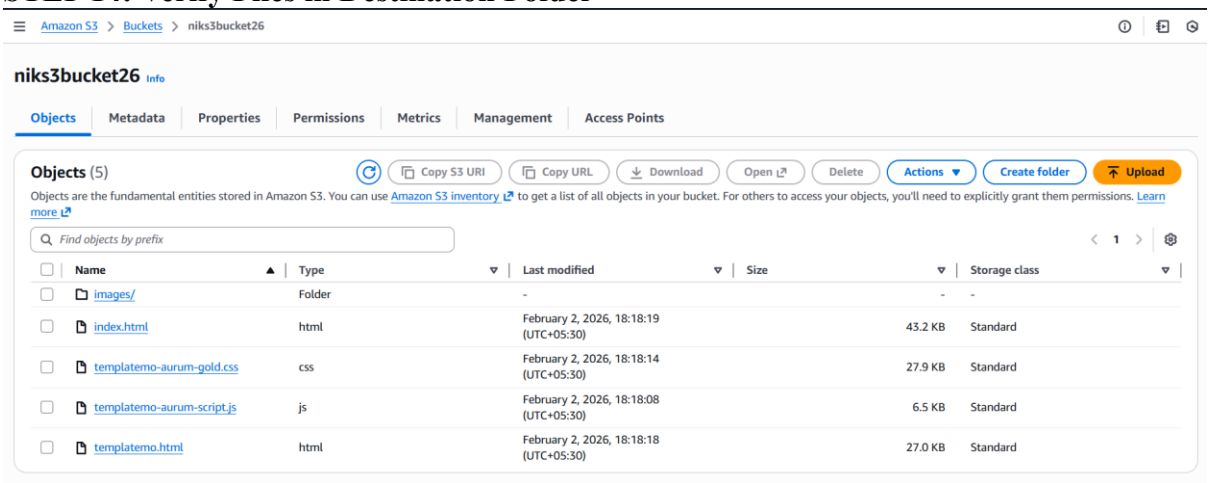

**STEP 12: Click Move Option**
After selecting the files, click on the **Actions** button and choose **Move**. This opens the Move objects window. In the **Move** window, keep **Destination type** as *General purpose bucket*.



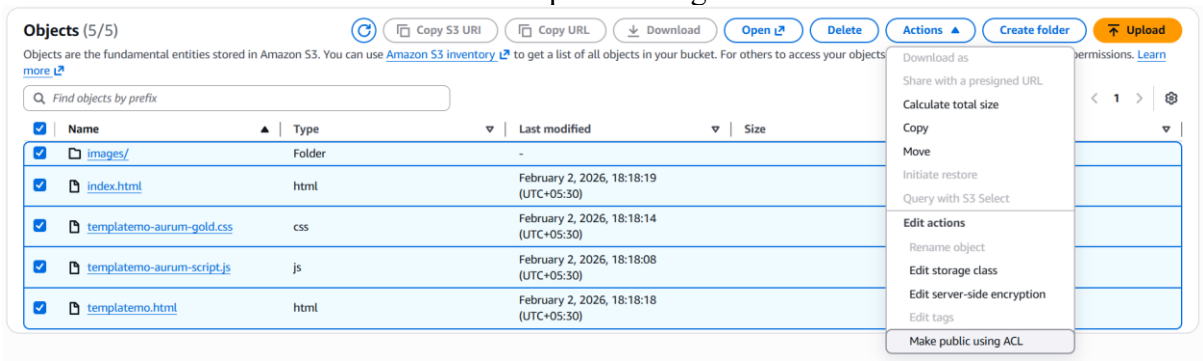
**STEP 13: Choose Destination Bucket and Folder**
Click on **Browse S3**, select the target bucket (for example: *niks3bucket26*), and choose the destination folder where the website files should be placed.



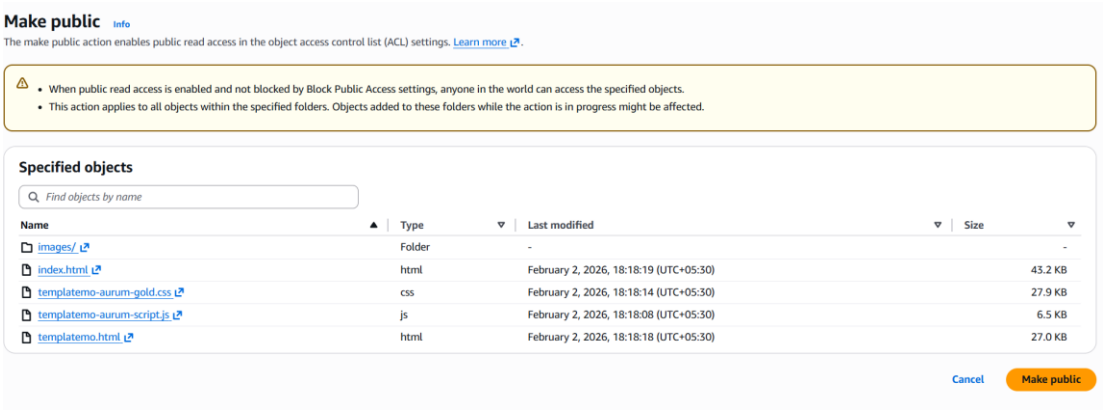

**STEP 14: Verify Files in Destination Folder**



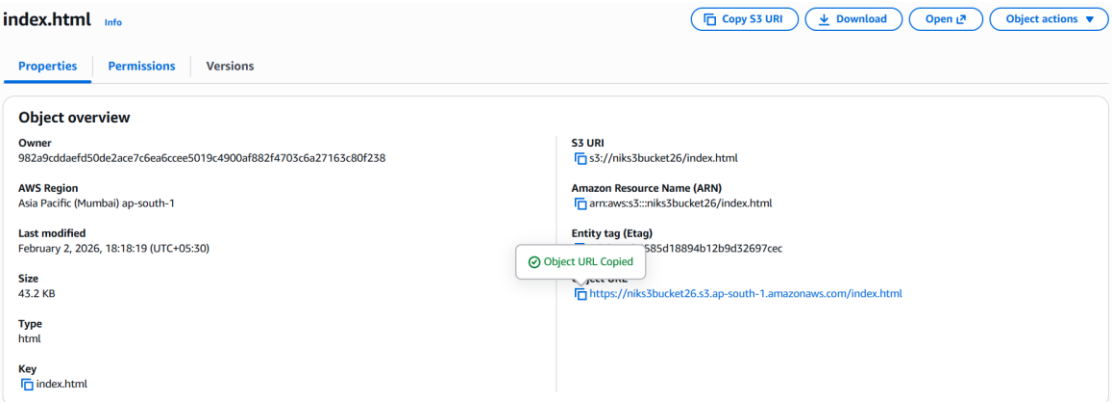

**STEP 15 : process of making uploaded website files public in the S3 bucket.**
Select all website files -> Actions ->Make public using ACL

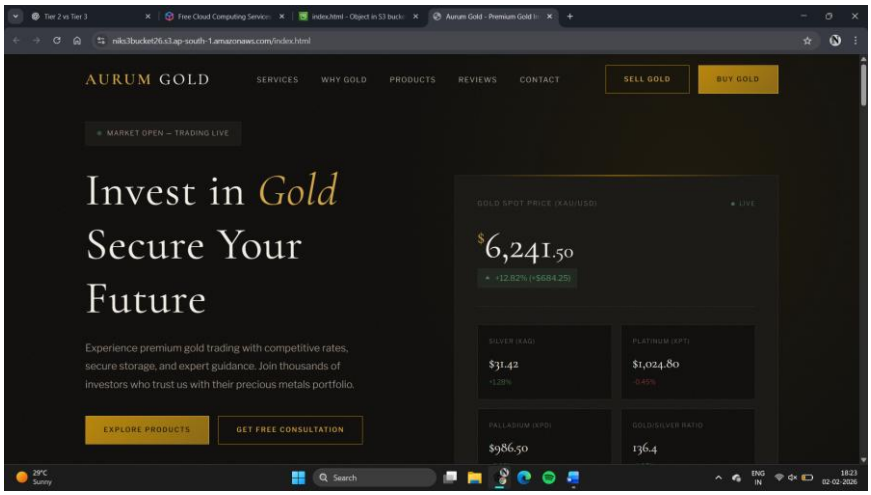


**STEP 16: Make Files Public**
Click on the **Make public** button to grant public read access to the selected objects.




**STEP 16: Open index.html File in S3**
Copy Object URL
From the object overview section, copy the **Object URL** provided for the index.html file.




**STEP 17: Verify Website Output**

**Open Website Using Object URL**
Paste the copied **Object URL** into a web browser
Verify that the website loads successfully in the browser, confirming that the static website has been deployed correctly using Amazon S3.